

Applied A.I. Solutions

Foundations of Data Management

Professor

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TOPICAL OUTLINE

- Week 1
1. Introduction to Data Management
 2. Data Governance
 3. Reference and Master Data Management,
 4. Metadata Management

- Week 2
1. Data Architecture
 2. Data Modeling and Design
 3. Data Integration and Interoperability
 4. Data Storage and Operations

- Week 3
1. Data Quality
 2. Document and Content Management
 3. Data Security
 4. Data Privacy

DATA MANAGEMENT INTRODUCTION

Introduction

By understanding and applying the principles of data management, AI professionals can build more **robust, reliable, and effective AI systems**.

- Data Quality and Integrity: Effective data management **ensures that the data used in AI models is accurate, consistent, and reliable**.
- Data Accessibility and Organization: Proper data management makes it **easier to locate, access, and use data efficiently**. This is essential for training AI models, which often require large datasets

Introduction - cont'd

- Data Security and Compliance: Managing data properly helps **protect sensitive information** and **ensures compliance** with relevant laws and regulations.
- Integration and Interoperability: AI systems often need to **integrate data from various sources**. Effective data management facilitates this integration, ensuring that data from different systems can be combined and used effectively
- Improving AI Performance: Good data management practices, such as data cleaning and preprocessing, are **essential for improving the performance of AI models**.

Essential Concepts – Data definition

- **Data** are discrete, **objective** facts or observations, which are unorganised and unprocessed, and do not convey any specific meaning or value because it is without context and interpretation.²
- **Information** is data that have been **contextualized** and shaped into a form that is meaningful and adds value to the understanding of a subject and useful to human beings.²

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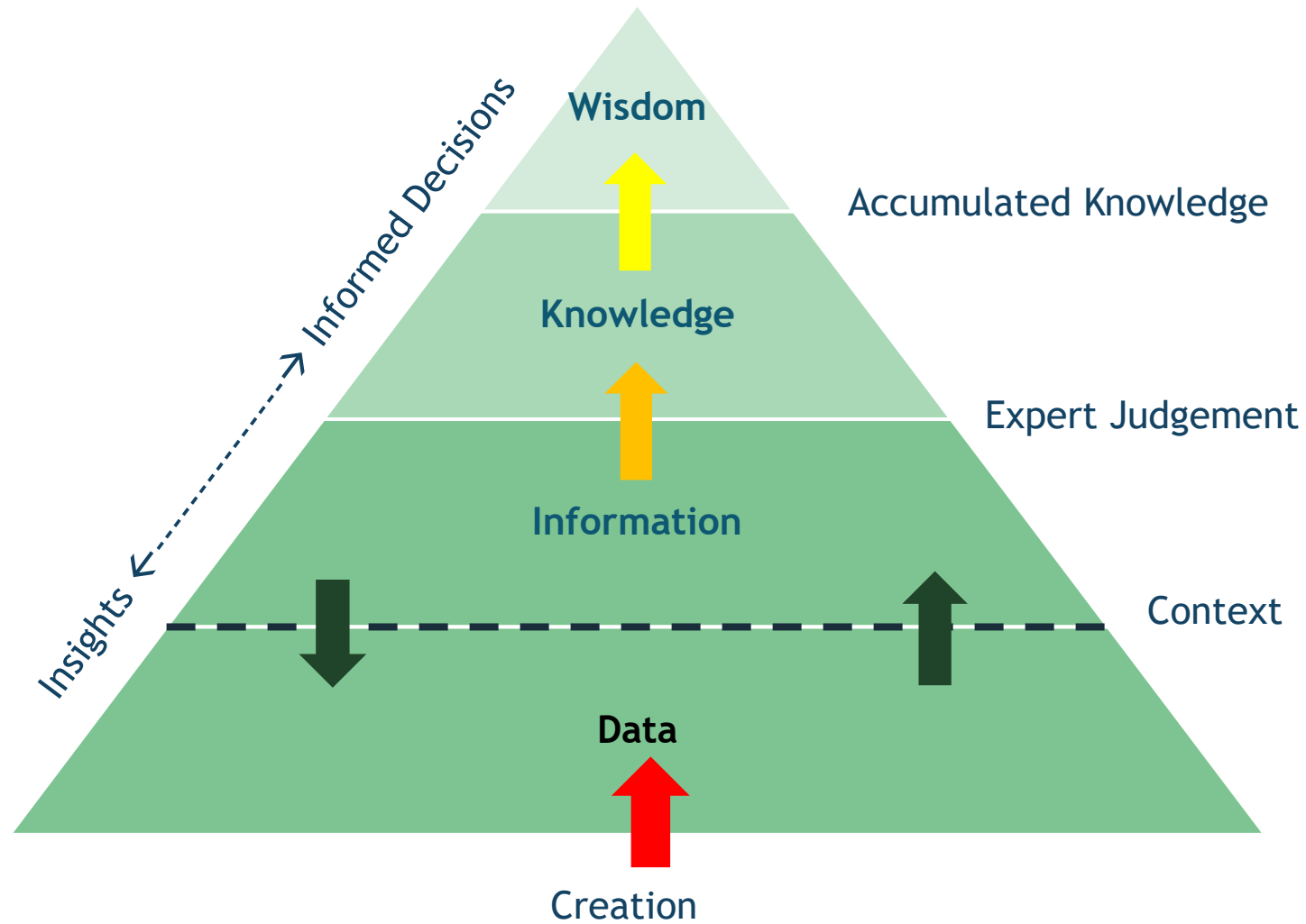
² Laudon and Laudon, 2006, Turban et al., 2005, Boddy et al., 2005. (Laudon and Laudon, 2006, p. 13). (Awad and Ghaziri, 2004, Chaffey and Wood, 2005, Pearlson and Saunders, 2004, Bocij et al., 2003). (Awad and Ghaziri, 2004, p. 36). (Jessup and Valacich, 2003, Bocij et al., 2003, Groff and Jones, 2003). (Chaffey and Wood, 2005, p. 233).

Essential Concepts – cont'd

- **Knowledge** is information to which is **added** expert opinion, skills, and **experience**, to result in an asset which can be used for informed decision-making and performance improvement.²
- **Wisdom** is the **accumulated knowledge**, the highest level of abstraction and the ability to act critically or practically in any given situation.²

² Laudon and Laudon, 2006, Turban et al., 2005, Boddy et al., 2005. (Laudon and Laudon, 2006, p. 13). (Awad and Ghaziri, 2004, Chaffey and Wood, 2005, Pearlson and Saunders, 2004, Bocij et al., 2003). (Awad and Ghaziri, 2004, p. 36). (Jessup and Valacich, 2003, Bocij et al., 2003, Groff and Jones, 2003). (Chaffey and Wood, 2005, p. 233).

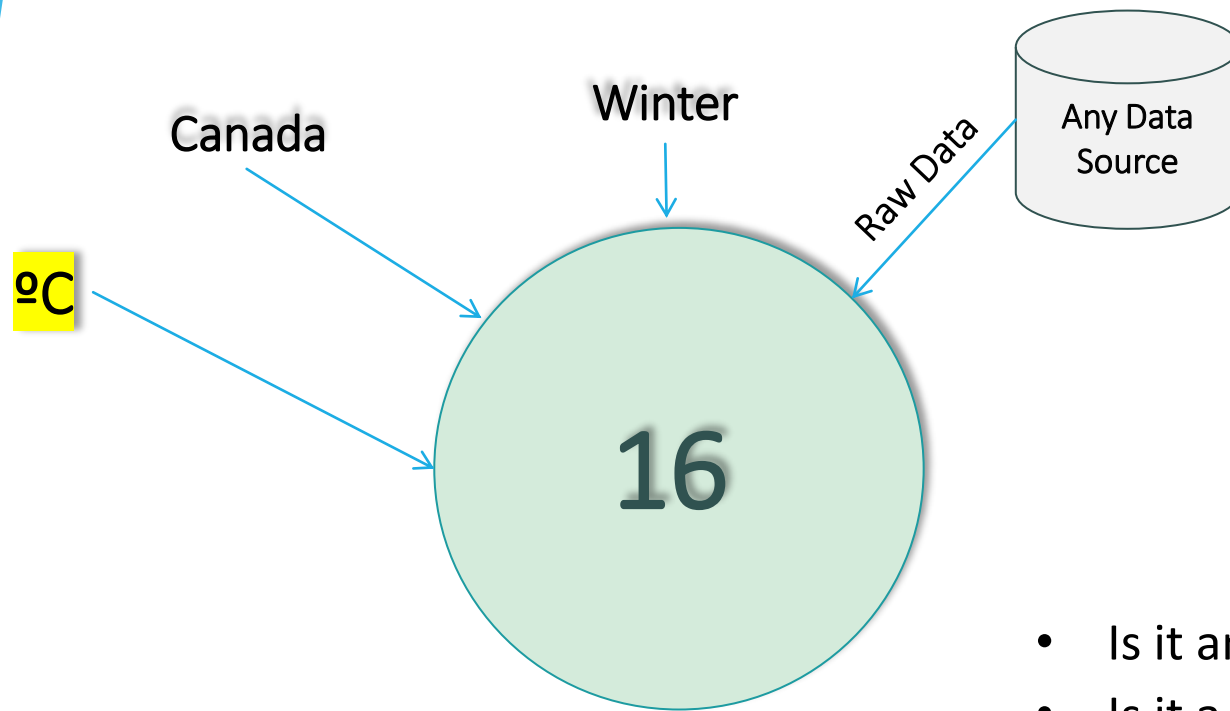
The DIKW Hierarchy



The DIKW Hierarchy - Rowley, 2007, p. 163

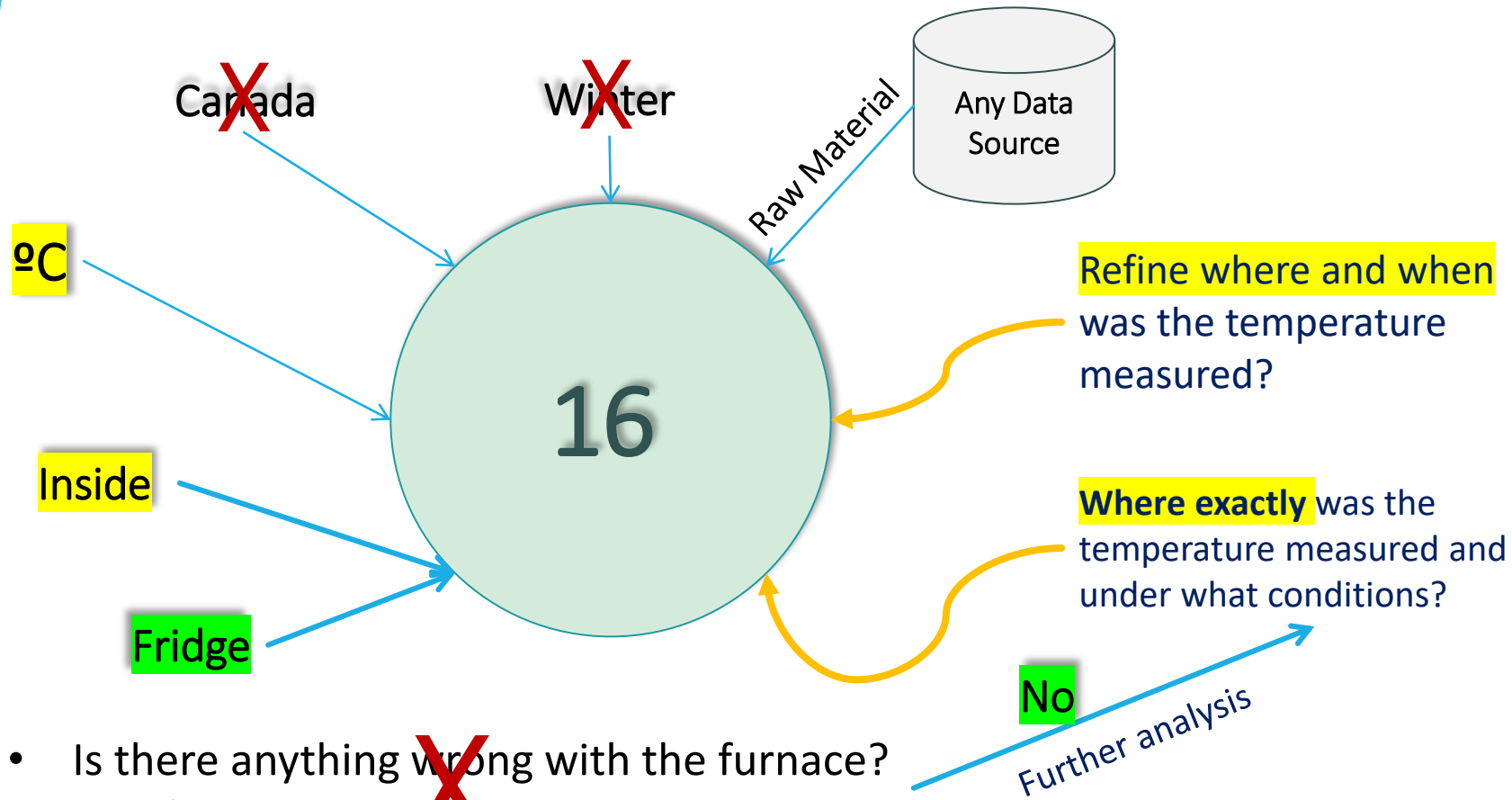
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What assumptions can you make? Insights about what did we got? What did we find?



- Is it an age?
- Is it a distance?
- Is it a weight?
- Is it a temperature?
- Which is the measurement unit?
- Where was it measured?
- When was it measured?

What assumptions can you make? Insights about what did we got? What did we find?



- Is there anything wrong with the furnace?~~X~~
- Are there any windows open?~~X~~

Yes
Too~~X~~cold!!!

✓ Is there anything wrong with the fridge?

→ Too warm!!!

- Data does not simple exists. **Data must be created.** It takes knowledge to create data
- **Data and Information are intertwined with, and dependent of each other.** Data is a form of information and information is a form of data
- **In DMBOK, the terms are used interchangeably**

Introduction - cont'd

Data Management is the development, execution, and supervision of plans, policies, programs, and practices that deliver, control, protect, and enhance the **value of data and information assets** throughout their data lifecycles.

Data Management Principles

- Data is an **asset** with unique properties
- Data value should be expressed in **economic terms**
- Managing data means managing the **quality** of data
- Using **context** to manage data
- Understanding the **information needs** of the **business** and its **stakeholders**

Data Management Principles - cont'd

- Capturing, storing, protecting, and ensuring the integrity of data assets
- Ensuring the quality of data and information
- Ensuring the privacy and confidentiality of data
- Ensuring data can be used effectively to add value to the business
- Preventing unauthorized access to information

Data Valuation

- **Value** is the difference between the cost of a product or service and the benefit(s) derived from that product or service.
- Data is **unique** to each organization and thus data valuation is not standardized.

Costs of data when it can be measured

- Collecting, storing data
- Data loss impact
- Risks
- Data improvement

Benefits of valuating data

- Identify higher quality
- What data could be sold for
- What competitor would pay
- Expected revenue from innovative uses of data

Valuation challenge and risks to consider

- Data is contextual (bias)
- Temporal, dynamic
- Type of data (velocity, veracity, variety, volume)

Impact of having low-quality data

- Scrap and rework
- Inefficiencies, low productivity
- Compliance costs, fines
- Opportunity costs
- Low job, client satisfaction

Impact of having high-quality data

- Improved customer experience
- Higher productivity
- Reduced risks
- Increased revenue
- Competitive advantage

Data Lifecycle

Data lifecycle is based on the product / service lifecycle.

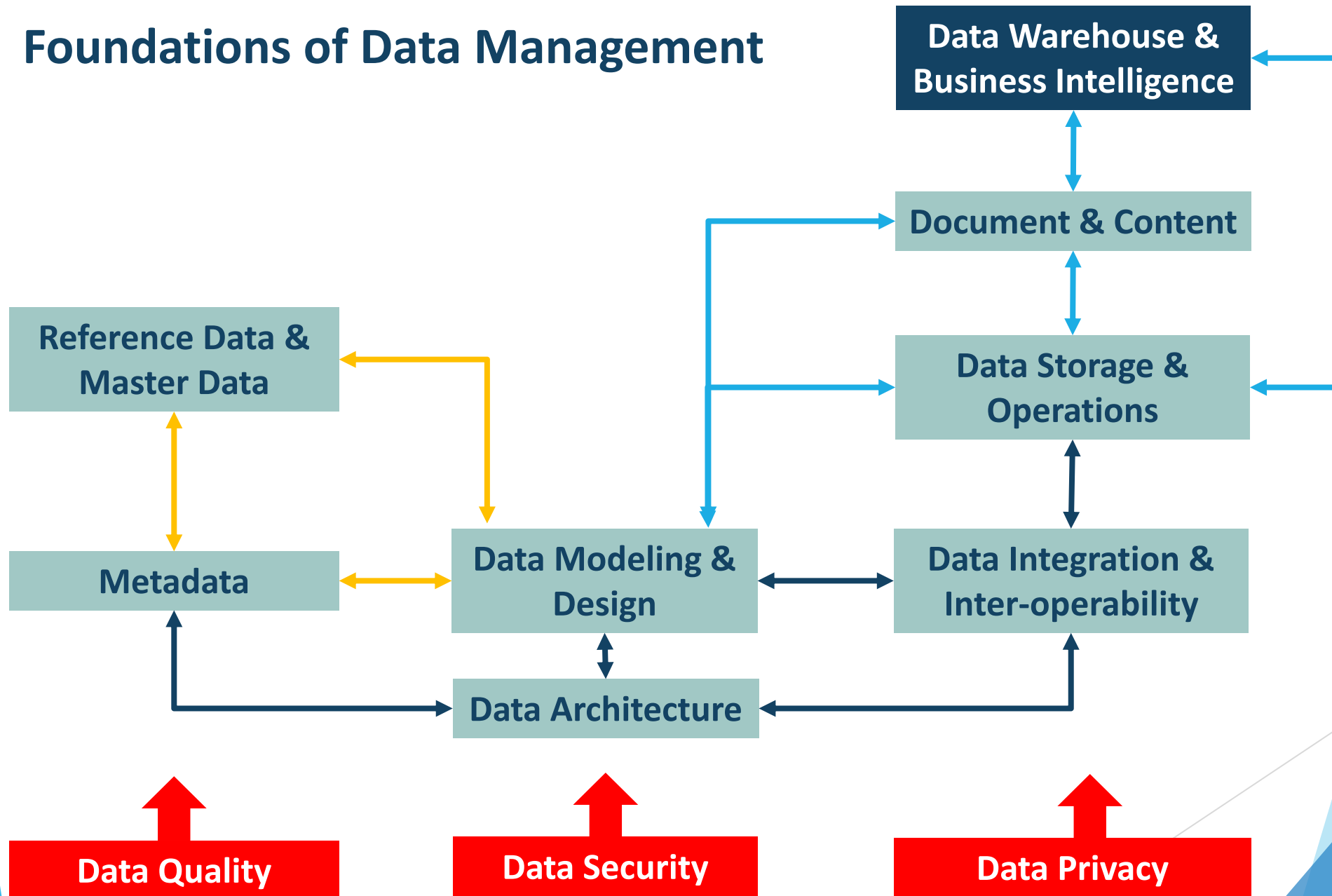
Lifecycle includes:

- Creation and usage
- Quality management
- Context / Metadata management
- Security management
- Critical data identification
- Data risk management

Data Management Framework

- A functional DM **framework** helps a data-driven organization to apply Data Management concepts, such as the use of a common vocabulary and serve as the **fundamental reference** guide for the Data Management Professional

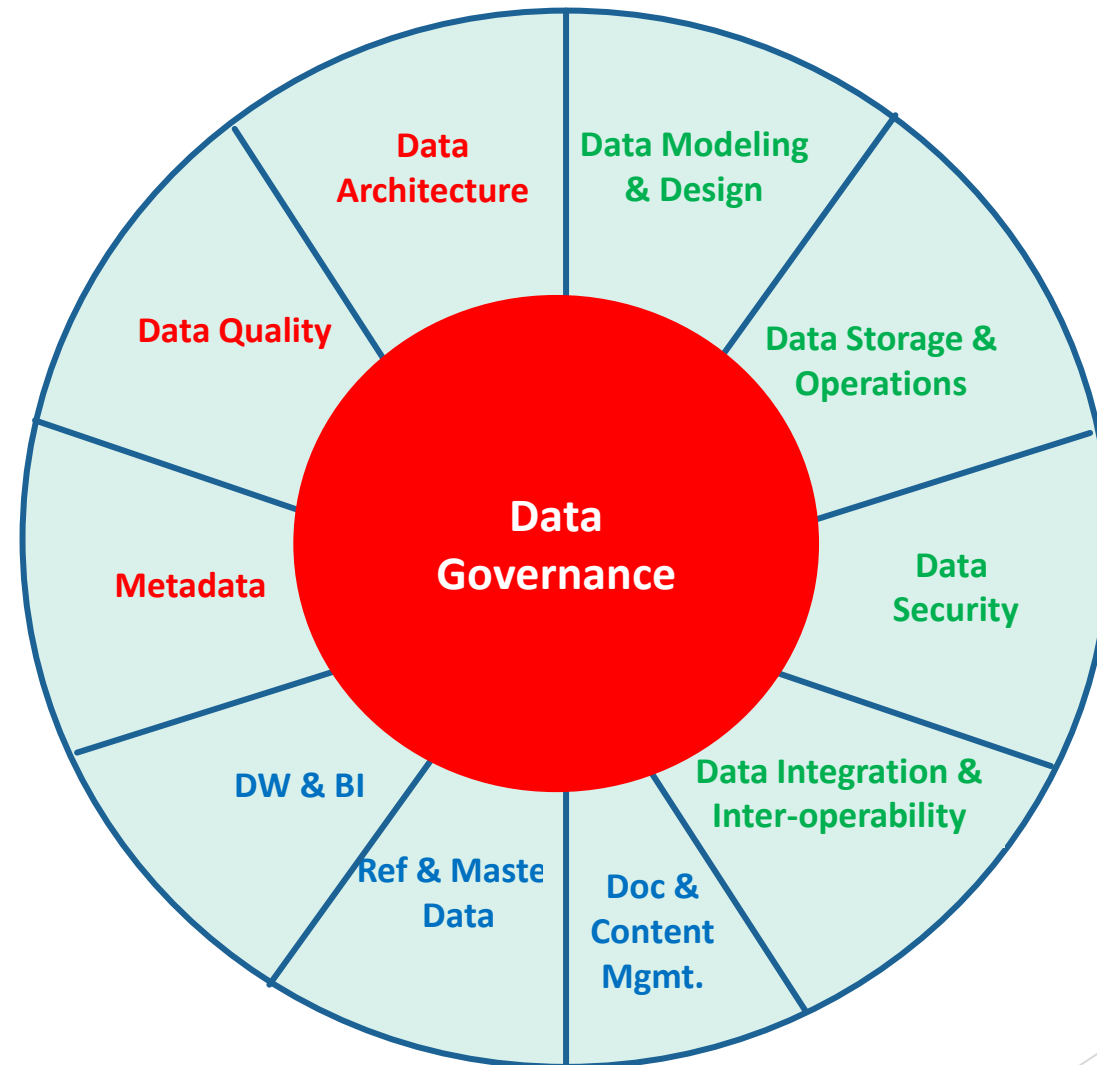
Foundations of Data Management



Models

1. Strategic Alignment Model
2. Amsterdam Information Model
3. **DAMA DMBOK framework – DAMA Wheel Model**
4. DMBOK Pyramid (Aiken)
5. DAMA Data Management framework evolved
6. DAMA Data Management Function framework
7. DAMA Wheel Model evolved

3. The DAMA Wheel



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